GG–I–271c
February 20, 1964

SUPERSEDING
Fed. Spec. GG-I-271b
May 25, 1954

FEDERAL SPECIFICATION

IGNITER, FRICTION, OXYACETYLENE-TORCH

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers hand-operated igniters (spark lighters), used for igniting gas welding and cutting torches. Underwater igniters are not covered.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 **Classification.**

1.2.1 *Types and styles.* Igniters shall be of the following types and styles, as specified (see 6.1):

Type I.—Wire frame, single flint.
Style A.—Round file, hooded.
Style B.—Flat file, hooded.
Style C.—Flat file, open.

Type II.—Flat frame, 3 flints, round file, hooded.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5120

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2.3 Other publications. The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

National Bureau of Standards Publication:

Handbood H28—Screw-Thread Standards for Federal Services.

(Copies may be obtained upon application to the Superintendent of Documents, Government Printing Office, Washington, D. C., 20402. Price may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Material. Material used in the manufacture of igniters shall be such as to produce igniters conforming to the requirements hereinafter specified.

3.2 Identification marking. Each igniter shall be marked in a permanent and legible manner with the manufacturer's name, or with a trademark of such known character that the source of manufacture may be readily determined.

3.3 Design and construction. Igniters shall be of a design consistent with the best commercial practices, and shall be so constructed that the parts will not become detached during normal use. In order to minimize chipping and breaking of flints, means shall be provided to prevent the flint from striking the hood or frame at either end of the file. Flint holders shall be of durable metal, and shall hold the flints securely without developing looseness during the useful life of the flint.

3.4 Frames. Frames shall be suitably shaped to facilitate spring action so that with normal hand pressure the flint end of the movable arm will travel lengthwise of the file and return when released.

3.5 Flints. The flints shall be of a pyrophoric alloy substance of adequate strength to withstand severe frictional movement across the file surface without chipping or breaking. The flints shall have a nominal diameter of 1/8 or 5/32 inch, and the frictional movement across the file surface shall produce a spark of sufficient intensity to ignite the gases used in welding and cutting torches. The flints shall be covered with a high grade waterproof paint or otherwise properly treated to minimize deterioration in storage.

3.6 Files. Files shall be of steel, heat treated to show a minimum hardness of 64 on the Rockwell C scale or case hardened to show a minimum hardness of 78 on the 30 N scale. Files shall be approximately 1 inch long and shall be securely attached to the fixed arm of the frame.

3.7 Finish. Unless otherwise specified in the contract or order (see 6.1), the surface finish shall be that which is normally produced by the best standard practices on the types and styles of igniters covered by this specification.

3.8 Spring compression. The movable arm of the frame shall be capable of passing the test specified in 4.3.1.

3.9 Flint igniting life. Flints shall be such as to pass the test specified in 4.3.2.

3.10 Type I, style A, wire frame, single flint, round file, hooded. Style A igniters shall consist essentially of a frame, flint, file, flint holder, and a hood.

3.10.1 *Frame.* Frames shall be of spring tempered wire having a minimum diameter of 1/8 inch. The end of the movable arm shall have at least 5 and no more than 7 threads, size number 5-40 NC of class 2 fit in accordance with National Bureau of Standards Handbook H28. The threaded end shall be bent so that the flint will be held approximately perpendicular to the file surface and shall otherwise conform to 3.4.

3.10.2 *Flint.* Flints shall conform to 3.5 and 3.9.

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3.10.3 *File.* The file shall conform to 3.6 except that the file shall be cylindrical in shape and rotate on the fixed arm of the frame.

3.10.4 *Flint holder.* Flint holders shall be similar to a metal sleeve with approximate dimensions of 3/16 inch outside diameter, 9/32 inch long, and a knurled section 1/8 inch long. One end of the holder shall have a minimum of 5 internal threads, size number 5-40 NC of class 2 fit in accordance with National Bureau of Standards Handbook H28.

3.10.5 *Hood.* Hood shall be of suitable metal, securely attached to the fixed arm of the frame, and partially surround the file without interfering with the normal operation of the movable arm.

3.10.6 Type I, style A igniter shall be similar to figure 1.

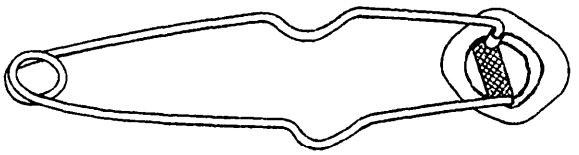


FIGURE 1.—Type I, style A wire-frame igniter, single-flint, round-file, hooded.

3.11 Type I, style B, wire frame, single-flint, flat-file, hooded. Style B igniters shall conform to style A, except that the file shall be of rectangular shape, and shall be similar to figure 2.

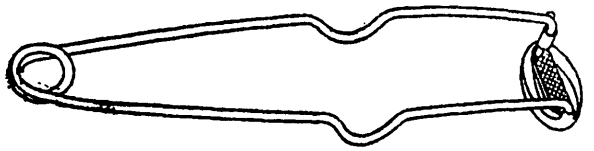


FIGURE 2.—Type I, style B wire-frame igniter, single-flint, flat-file, hooded.

3.12 Type I, style C, wire frame, single-flint, flat file, open. Style C igniters shall conform to style A, except that the file shall be of rectangular shape, and shall be furnished without a hood, and shall be similar to figure 3.

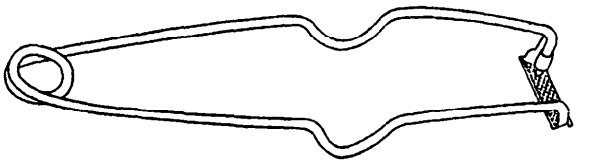


FIGURE 3.—Type I, style C wire-frame igniter, single-flint, flat-file, open.

3.13 Type II, flat frame, 3 flints, round file, hooded. Type II igniters shall consist essentially of a frame, flints, file, flint holder, and a hood.

3.13.1 *Frame.* The frame shall be of flat steel providing adequate sprint action of the movable arm when compressed and released. The movable arm shall provide means for securely attaching and adjusting the 3-flint holder. The fixed arm of the frame may be extended around the file and serve as part of the hood, or other suitable means provided for securely attaching a hood to the fixed arm.

3.13.2 *Flints.* Flints shall conform to 3.5 and 3.9.

3.13.3 *File.* The file shall conform to 3.6 except that the file shall be cylindrical in shape and rotate on a suitable shaft, which shall be securely fastened to the fixed arm of the frame.

3.13.4 *Flint holder.* Flint holder for type II igniters shall be designed to hold three individual flints with adequate means for proper positioning of one flint at a time over the file surface. Means shall be provided for readily replacing the flint holders on the movable arm of the frame.

3.13.5 *Hood.* Hoods shall conform to 3.10.5.

3.13.6 Type II igniter shall be similar to figure 4.

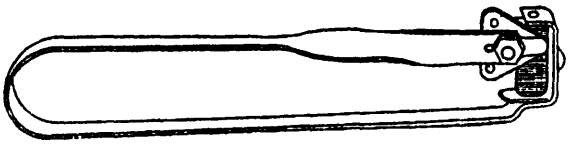


FIGURE 4.—Type II, flat-frame igniter, 3 flints, round file, hooded.

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3.14 Flint renewals. Flints, whether ordered at the same time with igniters or separated, shall be furnished assembled in the applicable flint holders, and shall conform to all the requirements herein (see 6.1.1).

3.15 Repair parts. The successful bidders supplying igniters covered by this specification, shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances and clearances, in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification.

3.16 Workmanship. Workmanship shall be first class throughout and free from defects which may impair the durability and serviceability of the finished product.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.2.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2.2 *Inspection of the end item.* Inspection of the igniters shall be performed as specified in the following paragraphs. A lot shall consist of igniters of the same type and style. The sample unit shall be one completely fabricated and assembled igniter.

4.2.2.1 *Visual examination.* Each sample unit shall be visually examined for the defects set forth in table I. The inspection level shall be level II, with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in defects per hundred units.

TABLE I.—*Classification of defects*

Examine	Defect	Major	Minor
Finish	Not finished in accordance with best standard practices	X	
Design	Not type or style specified	X	
Construction and workmanship	Component loose, missing, bent, fractured or will not funtion as intended	X	
	Flint not prevented from striking hood or frame	X	
	Flints not covered with high grade waterproof paint	X	
	Frames not shaped to facilitate spring action	X	
	Files not securely attached to fixed arm	X	
	Threads missing, stripped or otherwise defective	X	
	Holder does not hold flint securely	X	
	Hood not securely attached or otherwise defective	X	
	Defects in workmanship impairing durability or serviceability	X	
Marking	Missing, incomplete or not legible		X

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4.2.2.2 *Dimensional examination.* Each sample unit shall be examined for dimensions specified in applicable paragraphs. Any departure therefrom shall constitute a defect. The inspection level shall be S-3 with an acceptable quality level of 4.0 defects per hundred units.

4.2.2.3 *Testing.* Each sample unit shall be tested as specified in 4.3.1 through 4.3.2.1. Any determination not within specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 1.5 defects per hundred units.

4.2.3 *Inspection of preparation for delivery.* Inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with MIL-H-15424.

4.3 Tests.

4.3.1 *Spring compression load test.* Each of the sample igniters shall be tested to determine the spring compression load in the following manner:

For type I igniters, 3- and 6-pound weights are required. For type II igniters, 6-1/2- and 9-1/2-pound weights are required. Clamp the fixed arm of the frame in a horizontal position with the movable arm directly above. To the applicable type of igniter attach the required minimum and maximum weights, one at a time, to the movable arm 1-1/2 inches from the flint end. The minimum weight required for each type shall not compress the movable arm to the end of its normal travel. The maximum weight required for each type shall compress the movable arm to the end of its normal travel. Any igniter in the sample failing to conform to the spring compression load test shall be rejected, and if the number of nonconforming igniters in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.2 *Flint igniting life test.* Each of the sample igniters and spare flints shall be tested to determine the flint igniting life in the following manner:

Attach a flint holder to an igniter and ignite a gas several times to determine suitability, after which:

(a) Make certain that there is no loose material in flint holder; then weigh; record as "Weight No. 1".
(b) Ignite flame several times until protective covering is worn and a good spark is obtained; then reweigh.
(c) Ignite flame 50 times and reweigh.
(d) Ignite flame again 50 times and reweigh; find the loss in weight due to each 50 ignitions. If results differ more than 2 milligrams, continue process of igniting and reweighing until reliable results are obtained. From data, calculate loss in weight per ignition.
(e) Cut off flint with nippers as close to flint holder as possible and ignite until flint holder comes in contact with metal abrasive; weigh remainder and record as final weight; subtract final weight from weight No. 1 to find weight of usable flint; find number of ignitions by the following formula:

$$\frac{\text{Weight of usable flint}}{\text{Weight loss per ignition}} = \text{number of ignitions.}$$

4.3.2.1 Each flint shall be capable of producing a minimum of 1,500 ignitions. All weighing shall be in milligrams and tenths thereof. If the number of nonconforming flints in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions of the various levels of packaging protection for supplies and equipment.)

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5.1 **Preservation, packaging, packing and marking.** Preservation, packaging and packing shall be level A, B, or C, as specified (see 6.1).

5.1.1 *Levels A and B.* Igniters shall be preserved, packaged and packed in accordance with the applicable requirements of MIL-H-15424.

5.1.2 *Level C.* Igniters shall be cleaned, preserved and packaged in accordance with the supplier's commercial practice, and shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order (see 6.1), interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking required by the contract or order (see 6.1), exterior packages and exterior containers shall be marked in accordance with MIL-H-15424.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, and style (if applicable) of igniter required (see 1.2).
(*c*) Finish, if other than that ordinarily furnished by the manufacturer (see 3.7).
(*d*) Level of preservation, packaging, and packing required (see section 5).
(*e*) Marking, if different (see 5.2).

6.1.1 When ordering spare flints (renewals), specify whether for type I or type II igniters (see 3.14).

6.2 **Stocking requirements.** Due to deterioration of flints when stored, supply officers should keep stock at such a minimum that flints will not have been in storage longer than three months prior to issue.

6.3 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Lighters, Gas, Metal.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Lighters, Gas, Metal.
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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